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Probing Political Ideology in LLMs and VLMs: Latent Representations and Cross-Modal Generalization

By

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Abstract

Large language models (LLMs) encode rich internal representations of political ideology, mapped directly from the polarizing distributions of human internet data. Within the field of Computational Social Science, these models are increasingly utilized as "silicon samples" to simulate human populations, perform polling, and moderate content. However, it remains critically underexplored how these internal representations contribute to model decision-making, and how these latent semantic dimensions generalize across behavioral tasks and modalities. This thesis systematically investigates the causal role of learned ideological directions via inference-time interventions on attention head activations. First, I identify a liberal-conservative discourse dimension via linear probing using DW-NOMINATE scores and demonstrate its causal effects across text tasks, including political bias detection, simulated voting preferences, and ideological text neutralization. My findings reveal robust generalization in perception-based tasks, allowing algorithmic manipulation of the model's confirmation bias. However, I document stark asymmetries in simulated voting behaviors, exposing how post-training alignment guardrails (RLHF) fundamentally warp ideological representation spaces. Second, I extend this investigation into the multimodal frontier, illustrating that modern Vision-Language Models (VLMs) map visual political signifiers natively onto the exact identical liberal-conservative textual axis. I successfully execute zero-shot image token probing on Qwen3-VL utilizing the DW-NOMINATE text direction. Finally, I demonstrate cross-modal algorithmic steering on the generic LLaMA textual backbone of Janus-Pro, causally altering the narrative, demographic, and aesthetic generation of images without any explicit visual prompt engineering. This work highlights the profound risks of using ideologically biased foundation models for politically sensitive sociological tasks and underscores the necessity of managing latent political dimensions in unified multimodal systems.

Keywords: Computational Social Science; Political Ideology; Large Language Models; Vision-Language Models; Mechanistic Interpretability; Inference-Time Intervention

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1 Introduction

Large language models (LLMs) have exhibited an astonishing capacity to generate text reflecting a broad spectrum of ideological perspectives, including highly nuanced positions on polarizing political issues (Argyle et al., 2023; Kim et al., 2025; Le Mens & Gallego, 2025; Wu et al., 2023). In recent years, researchers within Computational Social Science (CSS) have increasingly turned to these foundation models as viable instruments for sociological

inquiry. Recent studies have revealed that LLMs can simulate the political views of specific U.S. demographic subgroups, lawmakers, and media outlets (Bernardelle et al., 2024; Santurkar et al., 2023). This capacity has sparked intense debate over the utility of LLMs as "silicon samples" capable of replacing or augmenting expensive human polling and focus groups.

Underpinning these capabilities is a profound geometric reality: these ideological stances are not merely stochastic emergent properties of next-token prediction algorithms. Instead, ideological orientations can often be linearly decoded directly from internal model activations using straightforward diagnostic probes (Kim et al., 2025; Park et al., 2024). This indicates that high-level abstract constructs like the *liberal–conservative* ideological spectrum are implicitly and robustly represented in discrete geometric regions of the model's high-dimensional activation space.

Despite these dramatic advances in mechanist interpretability, much of the existing research has focused on detecting and passively monitoring these linear ideological representations in either diagnostic contexts (Gurnee & Tegmark, 2023; Tigges et al., 2023) or isolated text generation contexts (Kim et al., 2025; Marks & Tegmark, 2023). Within the CSS community, there remains a critical gap in understanding whether these localized representations are functionally implicated in the broader decision-making behaviors of modern foundational models. If LLMs are to be trusted as analytical engines to code qualitative political data, detect bias, or simulate voter behavior, it is imperative to ascertain whether ideological dimensions identified via text probing are merely descriptive, inert features of the model's internal textual geometry, or if they exert direct causal influence on downstream social-scientific tasks.

Furthermore, these representational dynamics must be evaluated as models rapidly become multimodal. The transition from text-only LLMs to Vision-Language Models (VLMs) and Unified Language Models (ULMs) implies that the topological space where "conservatism" and "liberalism" reside is now integrating visual heuristics—how politicians dress, the aesthetics of rallies, the lighting of campaign ads, and the semiotics of rural versus urban landscapes. It remains entirely unknown how the single continuous latent space shared by text and vision in these new models contextualizes political ideology.

The overarching research questions driving this thesis are twofold: First, *how do latent ideological representations generalize across diverse, complex political reasoning tasks in text*? Second, *how does the linear dimension of political ideology discovered in textual syntax extend seamlessly to visual understanding and generation in Vision Language Models (VLMs)*?

To answer the first question, I move beyond descriptive probing to systematically test whether direct, programmatic interventions on the latent ideological dimension can steer

a model's performance. I extract this dimension utilizing the voting behavior patterns underlying DW-NOMINATE scores (Carroll et al., 2009), training a ridge-regression probe on the intermediate attention heads of prominent models (LLaMA-2, LLaMA-3.1, Qwen-2.5). I then apply Inference-Time Intervention (ITI) to perturb these activations during generation, causally testing the model's behavioral response across bias detection, simulated voting preferences, and the qualitative rewriting of political texts.

To answer the second question, I project this text-derived ideological representation onto visual inputs. Given that modern VLMs process vision by transforming image patches into the identical continuous token space shared with text, I probe the synthesized image tokens from Qwen3-VL utilizing my text-based DW-NOMINATE methodology. This reveals a deep cross-modal political alignment, allowing the text probe to map visual conservative/liberal biases in unstructured imagery. Further, I manipulate the generative visual outputs of the unified language model (ULM) Janus-Pro through algorithmic steering on its LLaMA backbone. By injecting the textual ideological vector into the image generation pathway, I demonstrate how political interventions fundamentally and stereotypically shift the aesthetic, demographic, and semiotic realities of the generated images.

My results illustrate a concerning asymmetry shaped by pretraining-induced alignment and human feedback reinforcement, confirming that models are not ideologically neutral, nor are they straightforwardly steerable in sensitive tasks like voting. By probing and causally manipulating the functional role of ideological directions across both text and vision, this work provides a technical tool-kit and a stark warning for CSS practitioners striving to audit, interpret, and depend upon multimodal foundation models.

2 Literature Review

This research bridges three distinct but converging domains: the measurement of political ideology within traditional social science, the emergence of ideological representations in neural networks, and the multimodal expansion of foundation models. By situating my methodology within these intersecting literatures, I establish the theoretical validity of linear latent interventions.

2.1 Measuring Political Ideology in CSS

The concept of political ideology is notoriously fluid, multidimensional, and highly context-dependent. Nevertheless, in American political science and Computational Social Science, it is traditionally operationalized along a single primary spatial dimension, typically the liberal–conservative axis. This spatial modeling assumes that legislators and concepts can be

placed on a geometric line capturing consistent partisan divides on economic redistribution, social policies, and foreign affairs (McCarty, 2016; Poole, 2005).

A cornerstone of such spatial measurements of ideology in the U.S. is the DW-NOMINATE score (Dynamic Weighted Nominal Three-step Estimation) (Carroll et al., 2009; Poole, 2005). DW-NOMINATE utilizes singular value decomposition on decades of congressional roll-call voting records to model lawmakers' behavior as a function of a low-dimensional space. The first dimension of DW-NOMINATE, representing the standard liberal-conservative divide, has proven incredibly robust in predicting broader policy alignments, polarization trends, and macro-political dynamics. While DW-NOMINATE is widely used due to its mathematical interpretability, it faces criticism from CSS scholars for relying exclusively on elite roll-call legislative data that may fail to capture dynamic issue-specific nuances, populism, or the multidimensional belief systems of the general public (McCarty, 2016).

Recently, CSS approaches have proposed complementary alternatives, including Bayesian item-response theory models (Caughey & Warshaw, 2015), campaign finance-based measures (Bonica, 2014), and natural language processing models that estimate ideal points directly from congressional speeches, tweets, or party manifestos (Vafa et al., 2020). However, no single text-based method has achieved the widespread, standardized adoption of roll-call based metrics.

My work leverages DW-NOMINATE anchoring. While it is derived from congressional elites rather than the corpus of internet text LLMs are trained on, in the U.S. political landscape, DW-NOMINATE's liberal–conservative axis remains a reliable "first-order" heuristic. As CSS literature demonstrates, online discourse heavily references, mimics, and polarizes around the exact legislative divides quantified by DW-NOMINATE. Therefore, it is highly likely that LLMs, trained on trillions of tokens of internet chatter, news articles, and Wikipedia, have implicitly learned to arrange their internal representations of political orientation along a topological axis that strongly mimics the DW-NOMINATE continuum.

2.2 Ideological Representations in language Models

With the advent of massive autoregressive models, a vibrant literature has emerged assessing LLMs as subjects of sociological inquiry. These models are increasingly utilized in "silicon sampling" to simulate human-like political behavior and support downstream applications such as multi-agent polling (Dai et al., 2024). Early milestone studies demonstrated that LLMs could adopt detailed partisan personas and accurately reflect the nuanced ideological preferences of specific demographic subgroups when provided with specific prompting conditions (Argyle et al., 2023; Potter et al., 2024). Subsequent work demonstrated that LLMs possess structured, internally consistent attitudes across contentious policy domains such as abortion, immigration, and foreign military aid (O'Hagan & Schein, 2023; Wu et al., 2023),

enabling their use as debate agents (Costello et al., 2024) and elements in simulating social media echo chambers (Mou et al., 2024; Park et al., 2024; Törnberg et al., 2023).

Despite CSS enthusiasm for LLM simulations, a persistent concern revolves around latent bias. LLMs may encode internal ideological orientations that silently skew their reasoning and text generation in ways that are not directly transparent to the researcher. These latent biases pose profound risks to the integrity of social simulations. Critically, these underlying biases are frequently resilient to post-hoc safety alignment techniques, such as instruction fine-tuning or reinforcement learning from human feedback (RLHF) (Gupta et al., 2023). RLHF often teaches a model to provide a seemingly neutral or evasive response to sensitive political queries (a "safety refusal"), masking the fact that the underlying representation space remains highly skewed, potentially distorting reasoning in subtle, pervasive ways under complex persona conditioning.

2.3 Probing, Representation, and Inference-Time Intervention

To uncover these latent biases beneath the RLHF veneer, researchers rely on probing methodologies. Probes evaluate whether the internal activations of neural networks—the intermediate matrices generated during a forward pass—encode abstract, human-understandable concepts (Alain & Bengio, 2016; Belinkov, 2022). Linear probes are heavily favored in interpretability research; they operate under the strong hypothesis that critical semantic features correspond to linearly separable directions in the model's high-dimensional representation space (Mikolov et al., 2013; Park et al., 2024). Linear probing has successfully revealed that LLMs encode sentiment, temporal relationships, and spatial geography in discrete geometric directions (Gurnee & Tegmark, 2023; Tigges et al., 2023).

Moving beyond mere diagnostic observations, recent literature explores Inference-Time Intervention (ITI). Li et al. (2023) formalized methods for computing steering vectors and modifying specific activations during the forward pass to causally steer a model's output behavior, originally utilized to enhance "truthfulness." Marks and Tegmark (2023) introduced causal tracing to manipulate factual knowledge retrieval. In a major leap for political analysis, Kim et al. (2025) applied these ideas to ideological dimensions, mapping the liberal-conservative axis via linear probes and demonstrating that scaling these pre-trained political vectors during generation predictably steers model text leftward or rightward.

My thesis substantially expands upon Kim et al. by shifting the focus from mimicking political personas to evaluating how these mathematically defined vectors generalize across complex, objective CSS reasoning tasks, evaluating the entanglement of ideology with bias detection and text neutralization.

2.4 Multimodality, Visual Framing, and Vision-Language Models

The study of political ideology is inherently multimodal. Within political science and communication literature, visual framing is a foundational concept. Politicians meticulously curate their visual aesthetics—from the formality of their clothing and lighting to the demographic composition of the crowds standing behind them—to signal ideological affiliation to their base. CSS researchers routinely analyze images from platforms like Instagram to predict the political leaning of social media users, identifying visual heuristics (e.g., trucks, rural landscapes vs. urban protests, lattes) that robustly correlate with conservative or liberal identities.

However, the question of how these visual political heuristics are encoded within foundational AI models is entirely unexplored. Modern Vision-Language Models (VLMs) and Unified Language Models (ULMs) construct unified representations by breaking image pixels into patches, passing them through vision encoders (like ViT), and projecting them into the identical continuous token space used to process English text. Because these models are trained on massive paired image-text datasets (e.g., LAION), where images of rural landscapes are frequently captioned with conservative-leaning text, and urban protests with progressive text, I hypothesize that the model algebraically maps these visual elements onto the exact same latent ideological axis utilized for textual syntax.

Recent progress in autoregressive unified models, which generate both text and images natively using a single decoder architecture, allows me to study cross-modal activation steering. By porting explicit mathematical representations of text-based ideology into the visual generation pathways of models like Janus-Pro, I can directly uncover the algorithmic physiognomy and aesthetic biases lurking within unified artificial intelligence.

3 Data and Methodology

To investigate whether latent ideological representations exert causal influence across text and vision tasks, I designed a robust, reproducible pipeline consisting of dataset generation, linear probe training, Inference-Time Intervention (ITI), and multi-modal evaluation. I focus on open-weight foundation models to allow full access to attention head activations: LLaMA-2 7B, LLaMA-3.1 8B, Qwen-2.5 7B for text tasks, and Qwen3-VL alongside Janus-Pro for vision tasks.

3.1 Dataset Construction for Text Probing and Tasks

To execute both linear probing and downstream evaluations, I required structured datasets of political text paired with objective measures of ideology. I rely on the DW-NOMINATE

scores (Dimension 1) from the 116th U.S. Congress as my continuous ideological labels, normalized strictly to the range $y \in [-1, 1]$. To support the two distinct phases of my methodology, I constructed two separate datasets.

3.1.1 The Ideology Probing Dataset

To train the ideology probe, I generated a dataset utilizing all lawmakers from the 116th Congress following the methodology of Kim et al. (2025). For each politician, I used a standardized prompt template:

```
USER: Generate a statement by [NAME], a politician in the United States.
ASSISTANT: In 2019, [NAME] said that
```

The model's internal activations were captured immediately at the end of this prompt shell. This dataset was exclusively utilized to train the linear probe.

3.1.2 The Policy Statement Dataset

To evaluate the causal efficacy of the trained probe across downstream CSS tasks (bias detection, rewriting), I required a diverse, domain-specific dataset. I semi-randomly sampled 40 distinct lawmakers across the ideological spectrum and selected six salient policy domains (*Abortion, Border Wall construction, Transgender Bathroom access, Military Aid to Ukraine, Military Aid to Israel, and Humanitarian Aid to Gaza*).

For each (lawmaker, domain) pair, I utilized `gpt-4o-mini` with a tighter simulation prompt:

```
SYSTEM: You are a politically accurate simulation engine.
USER: You are [NAME], a U.S. politician from the 116th Congress.  Write
a realistic one-sentence statement that [NAME] might give regarding
[DOMAIN].
```

This yielded 240 distinct political statements, serving as the base contextual material for evaluating Inference-Time Interventions.

3.2 Activation Extraction and Linear Probe Formulation

I isolated the latent ideological dimension within the internal geometry of the LLMs through ridge-regularized linear probing. Following the framework of Kim et al. (2025), I specifically target the activations of the *attention heads* across all layers, rather than the Multi-Layer Perceptron (MLP) layers, as attention heads heavily mediate contextual reasoning and perspective-taking.

For each politician in the probing dataset, I process the standardized string i through the target model (e.g., LLaMA-2) and record the internal activation vector produced at the final sequence token (the position explicitly tasked with predicting the ideological completion).

Let $x_{\ell,h}^{(i)} \in \mathbb{R}^{d_{\text{head}}}$ denote the activation of attention head h in layer ℓ corresponding to input statement i. I seek to find a linear projection vector $\theta_{\ell,h}$ that maps this activation to the empirical DW-NOMINATE ideological score $y^{(i)}$. For every attention head independently, I fit a ridge regression model:

$$\hat{\theta}_{\ell,h} = \arg\min_{\theta_{\ell,h}} \sum_{i=1}^{N} \left(y^{(i)} - \theta_{\ell,h}^{\top} x_{\ell,h}^{(i)} \right)^2 + \lambda \|\theta_{\ell,h}\|_2^2 \tag{1}$$

The ridge penalty λ is crucial; it mitigates severe overfitting given that the dimensionality of the head activations d_{head} (e.g., 128) is often comparable to my sample size N.

The trained coefficients $\hat{\theta}_{\ell,h}$ represent the "ideological direction" for that specific attention head. To filter out noise, I calculate the coefficient of determination (R^2) for each probe on a validation set. I subsequently rank all $L \times H$ attention heads (where L is layers, H is heads per layer) based on their predictive power, identifying the top-k heads most responsible for managing political ideology within the model.

3.3 Inference-Time Intervention (ITI) Steering

To test causality, I must prove that manipulating activations along the discovered direction $\hat{\theta}$ alters the model's textual and logical judgements. During the autoregressive generation of responses for downstream tasks, I intercept the forward pass after the attention calculation but before the residual connection.

For the selected top-k attention heads, I inject the steering vector into the activation $x_{\ell,h}$:

$$x_{\ell,h}^{\text{steered}} = x_{\ell,h} + \alpha \cdot \sigma_{\ell,h} \cdot \hat{\theta}_{\ell,h} \tag{2}$$

Here, α is a scalar hyperparameter governing intervention magnitude, and $\sigma_{\ell,h}$ is the historically computed empirical standard deviation of activations at that specific head. This ensures the injected vector remains within a plausible magnitude relative to the model's natural geometric distribution.

By varying $\alpha \in \{-30, -20, -10, 0, 10, 20, 30\}$, I programmatically force the model to adopt a more intensive liberal viewpoint ($\alpha < 0$) or a more intensive conservative viewpoint ($\alpha > 0$). When $\alpha = 0$, the model operates naturally (the baseline).

3.4 Downstream CSS Text Tasks

I evaluate the representational effects of ITI steering across three specific CSS tasks, utilizing identical intervention parameters across models.

1. **Political Bias Detection:** I evaluate if internal ideological representations algorithmically shift the model's perception of third-party bias. I present the 240 statements generated earlier. Under varying levels of α, I ask the model: *"In the context of U.S. politics, is the following statement biased and leaning toward a liberal or conservative viewpoint? Statement: [STATEMENT]"*. Responses are deterministically parsed to numeric scores (Liberal = 2, Neutral = 0, Conservative = -2).

2. **Voting Preference Prediction:** Simulating populations requires models to accurately reflect choices. I prompt the model: *"As a [LIBERAL/CONSERVATIVE], who would you most likely vote for in the next U.S. presidential election: Joe Biden or Donald Trump?"*. I observe changing probability distributions over the two candidates across shifting α to probe behavioral rigidity.

3. **Bias Neutralization via Rewriting:** The model is instructed: *"Rewrite the following statement to make it politically neutral and free from bias: [STATEMENT]"* under varied steering constraints. This evaluates whether ideology modulates subtle stylistic and lexical moderation choices.

3.5 Image Token Probing in Vision Language Models

To interrogate whether the linear dimension of political ideology discovered in textual networks extends cross-modally, I leverage Qwen3-VL, a state-of-the-art vision-language architecture. Modern VLMs process vision by segmenting images into discrete patches, encoding them through a Vision Transformer (ViT), and utilizing an adapter module to project these visual features into the exact continuous sequence space as text, typically represented by a sequence of `<|image_pad|>` tokens.

I hypothesize that these synthesized image tokens can be probed natively utilizing my text-trained ideology probe. I provide the VLM with politically charged images (e.g., politicians, rallies) and generic uncaptioned imagery from Unsplash, accompanied by a benign textual prompt: *"Describe the contents of this image in one sentence."*

During the forward pass processing the `<|image_pad|>` tokens within the transformer decoder layers, I cache the attention head activations $x_{\text{image_patch}}$. I subsequently compute the ideological dot product utilizing my *text-based* DW-NOMINATE weights: $y_{\text{image_patch}} = \hat{\theta}_{\ell,h}^{\top} x_{\text{image_patch}}$. By aggregating these scores across key heads and mapping them back to their corresponding spatial pixel coordinates, I construct intense visual ideological heatmaps,

identifying exactly which visual heuristics the model algebraically correlates with conservatism or liberalism.

3.6 Cross-Modal Generative Steering in ULMs

The ultimate verification of multimodal latent concept alignment is executing active generative control across modalities. I test this on Janus-Pro, an autoregressive Unified Language Model. Unlike heavily partitioned diffusion models, Janus-Pro fundamentally unifies modalities by predicting both discrete linguistic text tokens and quantized visual tokens utilizing a single shared LLaMA backbone transformer.

Because the visual sequence generation relies upon the exact same causal decoder pathway as text generation, I can implement Inference-Time Intervention (ITI) while the model attempts to draw an image. I inject the liberal ($-\alpha$) or conservative ($+\alpha$) text-derived steering vectors directly into the model's top attention heads across every step of the visual token generation loop. By supplying a simple base text prompt (e.g., "A politician") alongside contrasting steering magnitudes, I causally measure how abstract textual ideology alters the generated visual aesthetics, demographics, and lighting.

4 Results: Ideological Generalization Across Textual Tasks

I applied causal interventions along the DW-NOMINATE ideological dimension across LLaMA-2 7B, LLaMA-3.1 8B, and Qwen-2.5 7B. I explicitly varied the intervention intensity $\alpha \in \{-30...30\}$ while modulating the top $k = 32$ attention heads.

4.1 Generalization to Political Bias Detection

CSS researchers frequently use LLMs to classify massive text corpora for political bias. My results indicate that a model's capacity to categorize bias is deeply tethered to its spatial ideological orientation, which can be computationally overridden.

As depicted in Figure 1, when LLaMA-2 is steered heavily toward the political left ($\alpha = -30$, interpreting the text through a mathematically enforced progressive lens), the majority of neutral or lightly partisan statements are forcefully labeled as *Conservative*. Conversely, steering the model toward the right ($\alpha = 30$) causes the exact same input batch to be overwhelmingly classified as *Liberal*.

The pristine symmetric reversal proves that steering the model along a latent ideological direction algorithmically shifts its political set-point. To quantify this relationship, I calculated the Pearson correlation coefficient r between intervention strength α and the average parsed bias label. As shown in Table 1, both LLaMA-2 7B and Qwen-2.5 7B follow this

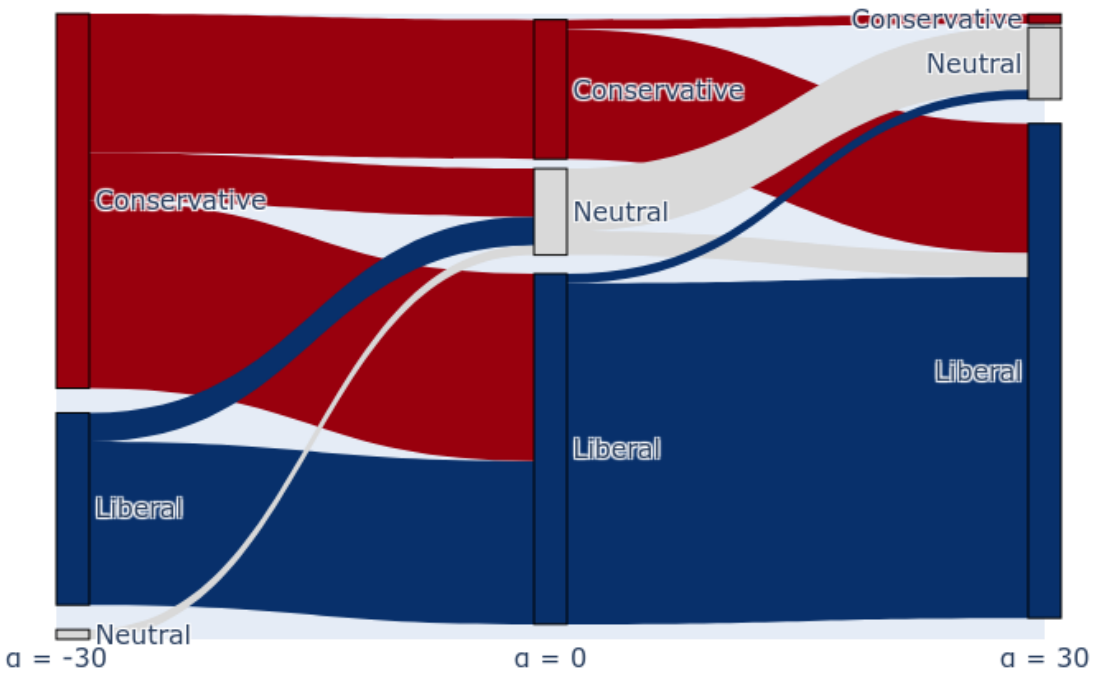

Figure 1: Sankey diagram showing transitions in political bias labels across intervention strengths ($\alpha = -30 \rightarrow 0 \rightarrow 30$) at $k = 32$ for LLaMA-2 7B. When the model is steered, it classifies identical texts relative to its new algorithmic frame.

k intervened	LLaMA-2 7B	LLaMA-3.1 8B	Qwen-2.5 7B
8	−0.98	0.88	−0.67
16	**−0.99**	0.43	0.35
32	−0.97	−0.55	**−0.99**
64	−0.72	**0.94**	−0.91
96	−0.85	0.83	−0.81

Table 1: Pearson correlation r between intervention strength α and average bias classification labels across models. Negative correlations indicate that steering toward liberal ($\alpha < 0$) causes the model to classify more statements as conservative, while positive correlations indicate the opposite.

deeply significant negative correlation pattern. The model evaluates third-party text relative to its own internal location on the manifold. In essence, the algorithmic equivalent of human confirmation bias functions perfectly: as the model's geographic position is pushed left, the entire "center" of the political spectrum moves right relative to its coordinate system.

Interestingly, LLaMA-3.1 exhibited reversed correlations compared to LLaMA-2 and Qwen. In LLaMA-3.1, steering the model liberal caused it to classify more statements as liberal. This indicates that LLaMA-3.1 collapsed the dimension of ideological representation and the dimension of bias-perception into identical semantic vectors during post-training fine-tuning, reflecting how different model families arrange identical data manifolds distinctly.

4.2 Asymmetries in Voting Preference Prediction

Simulating populations reliably requires models to faithfully reproduce behavioral choices. However, unlike the symmetry encountered in bias detection, manipulating behaviors for simulated ballot voting produced profound asymmetries and systemic distortions, mapped in Figure 2. To avoid degenerate outputs during generation, I explicitly restricted interventions during the voting tasks to $\alpha \in \{-20, -10, 0, 10, 20\}$ and $k \in \{16, 32, 64\}$, as extreme ± 30 magnitudes frequently produced totally incoherent or stubbornly unreadable completions.

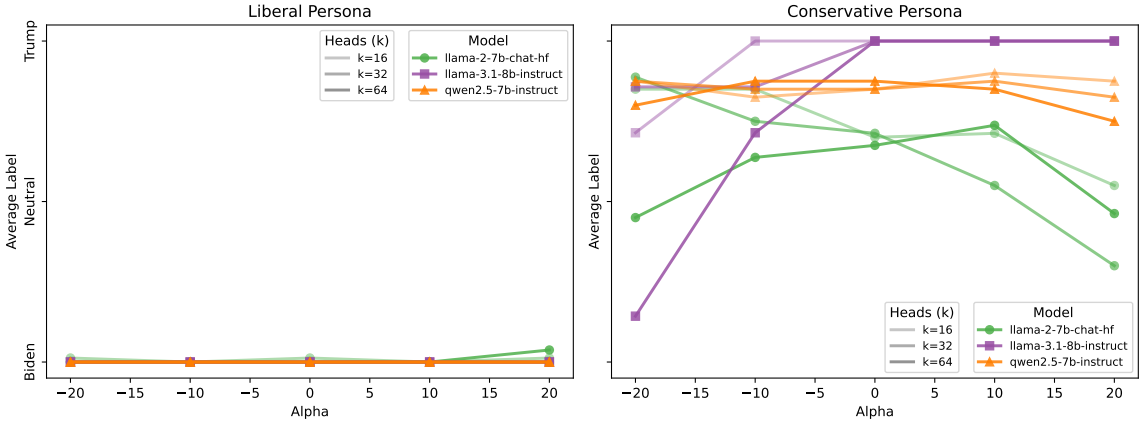

Figure 2: Average predicted voting preference (Biden = -1, Trump = 1) across intervention strengths α for several models. Liberal personas resist all manipulation, while conservative prompts yield erratic model-dependent results.

The **liberal persona** is overwhelmingly unsteerable across all three tested architectures. Irrespective of intervention magnitude α or the number of heads intervened k, the token outputs uniformly and stubbornly favored Joe Biden. The baseline liberal prior is seemingly

cemented mathematically.

The **conservative persona** displayed wildly erratic responses across architectures. The most striking observation is LLaMA-2 7B, which exhibited a highly counterintuitive dynamic: injecting heavy conservative steering vectors into the forward pass paradoxically increased the likelihood of predicting a Biden vote over a Trump vote.

This asymmetry is the profound and unintended shadow of Reinforcement Learning from Human Feedback (RLHF) and safety alignment. During alignment phases, human raters frequently penalize highly polarized, controversial, or aggressive conservative internet speech as "unsafe" or "toxic". Consequently, the LLM constructs intense safety guardrails immediately adjacent to conservative ideological vectors in the latent space. When my ITI mathematically shoves the activations deep into the conservative quadrant ($\alpha = 30$), the model's safety triggers interpreting the geometry as a violation. In order to avoid generating penalized text, the attention heads forcibly collapse the generation probabilities into benign, pre-approved progressive normative behaviors, creating an illusion of liberal voting preference out of defensive alignment. This finding serves as a stark warning against naively trusting LLMs for electoral behavior simulation in CSS.

An alternative, yet equally compelling explanation for this resistance to steering in behavioral simulation is that the explicit adoption of a partisan persona in the instruction prompt anchors the model's internal representations so rigidly that the ideological direction becomes innate and immutable. When the model is explicitly told, "As a liberal...", the attention heads may fully saturate the progressive direction on the activation manifold, rendering additional linear interventions functionally ineffective. This phenomenon closely mirrors recent findings by Lampinen et al. (2026), who demonstrated that when a model is instructed by the user to treat a fictional scenario as strict reality, its internal factual probes autonomously begin to activate on the fictional content. In my voting simulation, the explicit persona instruction may similarly "overwrite" the baseline ideological manifold, structurally locking the model into a fixed partisan reality that algebraically resists secondary steering vectors.

4.3 Rewriting and the Manipulation of Semantic Lexicons

I instructed models to neutralize ideologically charged statements under varied steering vectors. This evaluates whether latent ideology subtly modulates stylistic and lexical semantic choices beyond raw classification.

The baseline unsteered model ($\alpha = 0$) performs the task optimally, removing the partisan phrasing and providing balanced academic rhetoric. However, the $\alpha = -30$ injection aggressively overrides the instruction prompt to remain "neutral." As seen in Table 2, the leftward vector forces the mandatory adoption of modern progressive academic vocabulary

α	**Steer**	**Output Excerpt**
–	Original	"As we navigate... it is essential to respect individuals' privacy while also ensuring that all students feel safe..."
−30	Liberal	"...recognize the importance of respecting individuals' privacy... while also addressing the ongoing struggle for justice and equality in the face of systemic oppression and discrimination."
0	Neutral	"...strike a balance between respecting individuals' privacy and creating an inclusive environment..."
30	Conservative	"...consider the privacy of individuals... specific actions and preferences of individuals should be taken into account..." (incoherent continuation follows)

Table 2: Impact of latent ideology on neutrality rewriting. Heavy steering overrides the core instruction prompt, forcing the adoption of specific partisan vocabularies.

("systemic oppression", "ongoing struggle for justice"). The model conflates extreme liberal ideology with its natural baseline generative vocabulary for sociology.

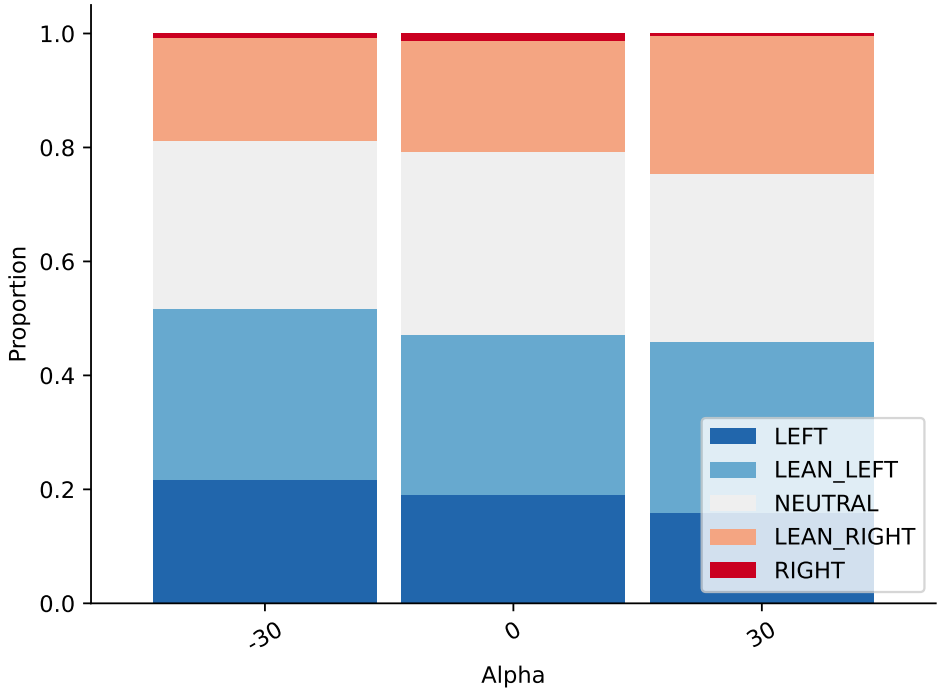

Figure 3: GPT-5 classification of the generated rewrites. High magnitude interventions destroy neutrality, polarizing the rewritten text along partisan lines.

Conversely, the $\alpha = 30$ high-magnitude conservative injection highlights "individual ac-

tions" but swiftly fragments into grammatical incoherence, mirroring the RLHF constraint conflict observed in the voting simulations. These findings demonstrate that latent linear interventions aggressively control abstract generative style and lexicon, completely decoupling the generated text from the explicit neutrality instruction of the system prompt.

5 Results: Ideological Generalization in Vision Models

Having established a robust algorithmic mapping of ideology that drives textual behavior, I cross the modality barrier to examine how multimodal architectures associate raw pixels and aesthetics with political structures.

5.1 Cross-Modal Alignment: Image Token Probing

When I applied my DW-NOMINATE *text probe* matrix multiplications directly against the sequence activations of the Qwen3-VL *image processing tokens*, the geometry demonstrated perfect alignment across modalities.

As the VLM generated a benign, single-sentence caption summarizing a photograph of a recognizable politician, my probe mapped rolling ideological scores across the spatial grid of visual tokens. The spatial regions encapsulating the figure's face distinctly and intensely triggered massive partisan activation weights precisely aligned with that politician's real-world ideology. The continuous array of pixel tokens registers along the identical mathematical axis as text.

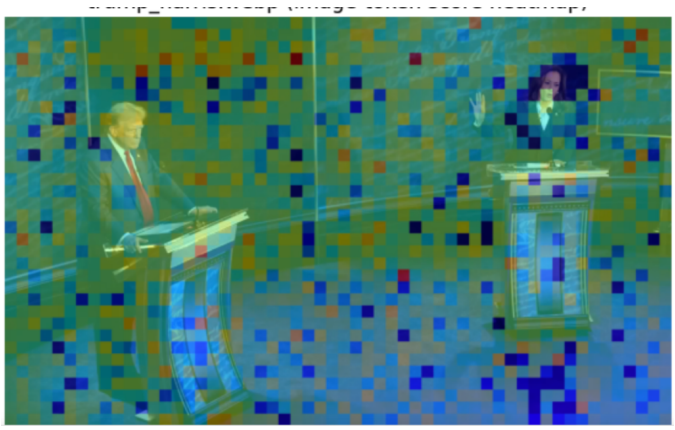

Figure 4: Visual ideology score heatmapping of an image patch on Qwen3-VL. The text-trained ideology probe flawlessly localizes the spatial presence of the political figure and assigns the correct partisan tensor weight, without taking any explicit multimodal retraining.

The faces of U.S. presidents, campaign trail podiums, and political signs light up fiercely when multiplied by the text vector, emphatically proving that foundational intelligence utilizes one unified spatial concept map to understand the semantics of "conservatism" whether reading a bill or looking at a photograph.

5.2 Algorithmic Physiognomy in Unlabeled Images

To push past explicit political photography, I employed the text ideology probe against a large corpus of 25,000 generic pictures from Unsplash to discover the ambient algorithmic geography of ideology.

This analysis revealed the VLM's structured ontological belief system regarding the aesthetic signifiers of human political affiliations.

- **Liberal Vector Triggers**: The model intrinsically associates dense urban environments, large crowds spanning diverse demographics, environmental protests, and progressive bohemian fashion heuristics massively with the liberal text vector.
- **Conservative Vector Triggers**: In rigid juxtaposition, the VLM associates wide-open rural landscapes (e.g., cornfields), heavy industrialized farming equipment (tractors), military pageantry, pickup trucks, and traditional Americana settings forcefully with the conservative text vector.

The foundational model has consumed the cultural stereotypes of Western political sociology from internet imagery. It quantitatively enforces the sociological divide between the liberal dense-urban sphere and conservative rural-industrial aesthetic natively within its most central decoder layers.

5.3 Steering Aesthetic Generation in Unified Language Models

The decisive evidence demonstrating the causal entwinement of text and vision modalities is my capacity to steer generated imagery utilizing text-vector arithmetic inside the Unified Language Model Janus-Pro.

By directly manipulating the forward sequence generation parameters utilizing my ITI text methodology—replacing natural attention activations with ideological additions—I systematically altered the visual scene generated by the generic prompt: *"A politician"*.

As shown in Figure 5, scaling α backward and forward triggers vast contextual shifts entirely consistent with the VLM heatmaps. Leftward progressive steering commands brighter natural daylight, dense demographically diverse outdoor environments, softer expressions, and casual open-collar approachability. Conversely, Rightward conservative steering enforces highly structured, symmetrical framing, solitary subjects isolated in opulent or wooden rooms, dark dramatic lighting, rigid posture, and severe formal suits.

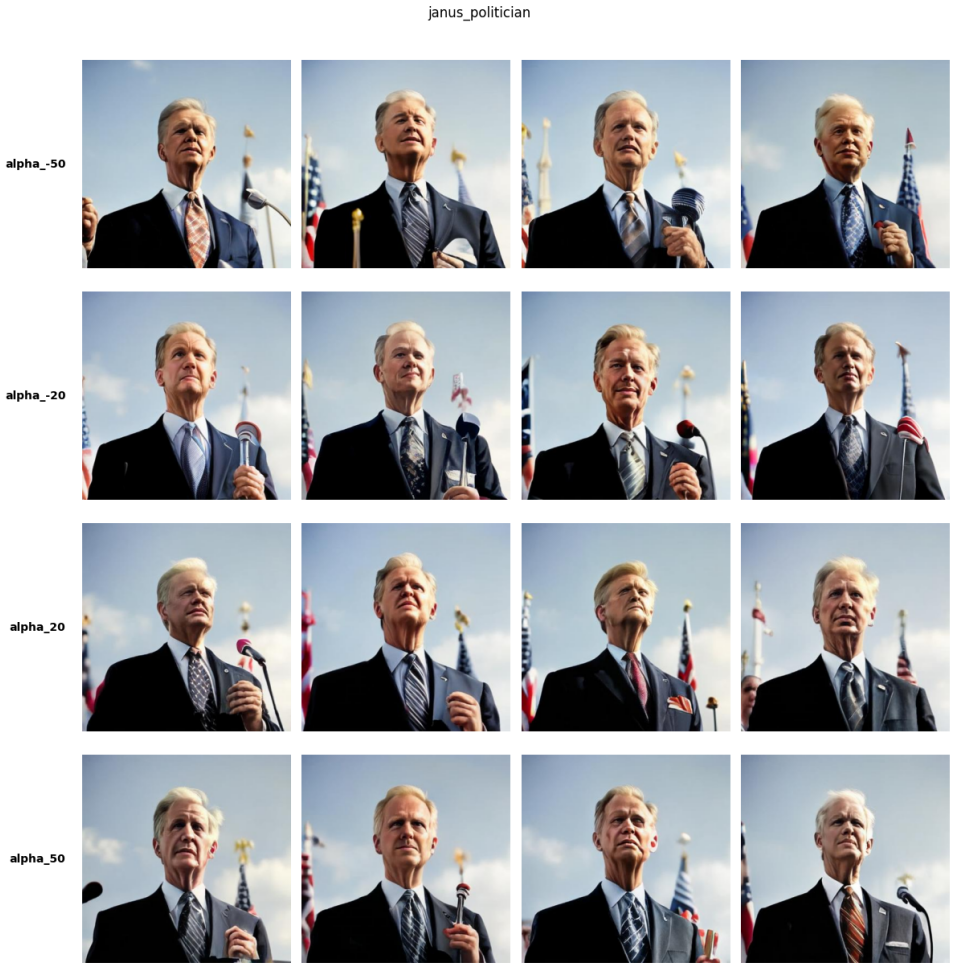

Figure 5: Generational output for the minimal prompt "A politician", steered algorithmically across the liberal-conservative axis. Textual linear math completely dictates the visual representation.

A solitary linear mathematical vector extracted from textual, legislative DW-NOMINATE data seamlessly and thoroughly rewrote the visual atmosphere, demographic implications, and artistic lighting of standard generations produced by the ULM framework.

6 Societal Implications for Computational Social Science

The empirical findings from this thesis carry deep methodological and ethical implications, particularly for researchers and technologists in Computational Social Science utilizing LLMs as proxies for human studies.

1. Algorithmic Physiognomy and Inherited Stereotypes

The discovery that VLMs and ULMs map specific fashion, demographic compositions, and geographic landscapes (e.g., rural vs. urban) directly onto biological political ideology vectors represents a terrifying reinforcement of algorithmic physiognomy. If a model intrinsically associates specific facial structures or agricultural environments as inherently right-wing, and urban density and diverse backgrounds as inherently left-wing, any automated analysis pipeline utilizing these models will suffer from massive systemic prejudice. When generating avatars, moderating photos, or tagging social media images, the foundation layers of artificial intelligence are currently poised to aggressively enforce mid-20th-century cultural caricatures.

2. The Illusion of Simulated Polling

My finding regarding the deep, unsteerable asymmetries forced by RLHF indicates that CSS researchers must view LLM behavior simulation with extreme skepticism. Models do not faithfully represent true distributions of nuanced human behavior; their ideological generation spaces are warped intentionally by safety guardrails. As evidenced by LLaMA-2 collapsing a conservative simulation request into a generic safety-aligned liberal response, utilizing these tools for predicting elections, gauging polarized public sentiment, or designing marketing messages risks harvesting hallucinations generated entirely by alignment-driven conformity.

3. Automated Bias in Next-Generation Media

The seamless capability to algorithmically steer a model's textual vocabulary (injecting "systemic oppression" automatically) and its visual aesthetic framing (shifting lighting from sunny crowds to dark, austere rooms) utilizing continuous mathematical vectors demonstrates an unprecedented surface for manipulative automation. As Unified Language Models lower the cost of high-definition multimedia propaganda to zero, malicious actors can simply apply a $+25\alpha$ vector to spin universally tailored, ideologically polarized synthetic media in real-time, locking audiences into multimodal echo chambers generated with subliminal aesthetic perfection.

7 Conclusion and Future Work

This thesis establishes that latent political ideological dimensions in foundation models are not merely descriptive emergent epiphenomena, but highly functional, cross-modal geometric pathways that can be extracted, diagnosed, and causally manipulated to shift system behaviors.

Generalizing ideological probing far beyond simple text persona imitation, I proved how steering algorithmically shifts the internal classification logic of models and radically manipulates semantic text generation. My novel multimodal investigations utilizing Qwen3-VL and Janus-Pro confirm that foundational visual intelligence intrinsically anchors upon textual ideological primitives to comprehend visual context, mapping legislative text voting data seamlessly onto photographs of faces and landscapes.

Moving forward, interpretability and CSS researchers must pioneer multi-dimensional steering methodologies to separate partisan identity from cultural aesthetics. How exactly do continuous image diffusion architectures specifically inherit these language vectors? Can researchers construct complex "Mixtures of Steering" designed to artificially synthesize entire voter demographic arrays for controlled societal simulations? As the boundary between text and unified visual intelligence disappears entirely, the advancement of robust safety protocols demands that I maintain precise, quantitative vigilance over the political steering wheels embedded within my foundational architectures.

References

Alain, G., & Bengio, Y. (2016). Understanding intermediate layers using linear classifier probes. https://doi.org/10.48550/ARXIV.1610.01644

Argyle, L. P., Busby, E. C., Fulda, N., Gubler, J. R., Rytting, C., & Wingate, D. (2023). Out of One, Many: Using Language Models to Simulate Human Samples. *Political Analysis*, *31*(3), 337–351. https://doi.org/10.1017/pan.2023.2

Belinkov, Y. (2022). Probing Classifiers: Promises, Shortcomings, and Advances. *Computational Linguistics*, *48*(1), 207–219. https://doi.org/10.1162/coli_a_00422

Bernardelle, P., Fröhling, L., Civelli, S., Lunardi, R., Roitero, K., & Demartini, G. (2024). Mapping and Influencing the Political Ideology of Large Language Models using Synthetic Personas. https://doi.org/10.48550/ARXIV.2412.14843

Bonica, A. (2014). Mapping the Ideological Marketplace. *American Journal of Political Science*, *58*(2), 367–386. https://doi.org/10.1111/ajps.12062

Carroll, R., Lewis, J. B., Lo, J., Poole, K. T., & Rosenthal, H. (2009). Measuring Bias and Uncertainty in DW-NOMINATE Ideal Point Estimates via the Parametric Bootstrap. *Political Analysis*, *17*(3), 261–275. https://doi.org/10.1093/pan/mpp005

Caughey, D., & Warshaw, C. (2015). Dynamic Estimation of Latent Opinion Using a Hierarchical Group-Level IRT Model. *Political Analysis*, *23*(2), 197–211. https://doi.org/10.1093/pan/mpu021

Costello, T. H., Pennycook, G., & Rand, D. G. (2024). Durably reducing conspiracy beliefs through dialogues with AI. *Science*, *385*(6714), eadq1814. https://doi.org/10.1126/science.adq1814

Dai, G., Zhang, W., Li, J., Yang, S., lbe, C. O., Rao, S., Caetano, A., & Sra, M. (2024). Artificial Leviathan: Exploring Social Evolution of LLM Agents Through the Lens of Hobbesian Social Contract Theory. https://doi.org/10.48550/ARXIV.2406.14373

Gupta, S., Shrivastava, V., Deshpande, A., Kalyan, A., Clark, P., Sabharwal, A., & Khot, T. (2023). Bias Runs Deep: Implicit Reasoning Biases in Persona-Assigned LLMs. https://doi.org/10.48550/ARXIV.2311.04892

Gurnee, W., & Tegmark, M. (2023). Language Models Represent Space and Time. https://doi.org/10.48550/ARXIV.2310.02207

Kim, J., Evans, J., & Schein, A. (2025). Linear Representations of Political Perspective Emerge in Large Language Models. https://doi.org/10.48550/ARXIV.2503.02080

Lampinen, A. K., Li, Y., Hosseini, E., Bhardwaj, S., & Shanahan, M. (2026, February). Linear representations in language models can change dramatically over a conversation. https://doi.org/10.48550/arXiv.2601.20834

Le Mens, G., & Gallego, A. (2025). Positioning Political Texts with Large Language Models by Asking and Averaging. *Political Analysis*, 1–9. https://doi.org/10.1017/pan.2024.29

Li, K., Patel, O., Viégas, F., Pfister, H., & Wattenberg, M. (2023). Inference-Time Intervention: Eliciting Truthful Answers from a Language Model. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, & S. Levine (Eds.), *Advances in Neural Information Processing Systems* (pp. 41451–41530, Vol. 36). Curran Associates, Inc.

Marks, S., & Tegmark, M. (2023). The Geometry of Truth: Emergent Linear Structure in Large Language Model Representations of True/False Datasets. https://doi.org/10.48550/ARXIV.2310.06824

McCarty, N. (2016). In Defense of DW-NOMINATE. *Studies in American Political Development*, *30*(2), 172–184. https://doi.org/10.1017/S0898588X16000110

Mikolov, T., Yih, W.-t., & Zweig, G. (2013). Linguistic regularities in continuous space word representations. In L. Vanderwende, H. Daumé III, & K. Kirchhoff (Eds.), *Proceedings of the 2013 conference of the north American chapter of the association for computational linguistics: Human language technologies* (pp. 746–751). Association for Computational Linguistics.

Mou, X., Wei, Z., & Huang, X. (2024). Unveiling the Truth and Facilitating Change: Towards Agent-based Large-scale Social Movement Simulation. *Findings of the Association for Computational Linguistics ACL 2024*, 4789–4809. https://doi.org/10.18653/v1/2024.findings-acl.285

O'Hagan, S., & Schein, A. (2023). Measurement in the Age of LLMs: An Application to Ideological Scaling. https://doi.org/10.48550/ARXIV.2312.09203

Park, K., Choe, Y. J., & Veitch, V. (2024). The linear representation hypothesis and the geometry of large language models. *Proceedings of the 41st International Conference on Machine Learning*.

Poole, K. T. (2005, April). *Spatial Models of Parliamentary Voting* (1st ed.). Cambridge University Press. https://doi.org/10.1017/CBO9780511614644

Potter, Y., Lai, S., Kim, J., Evans, J., & Song, D. (2024). Hidden Persuaders: LLMs' Political Leaning and Their Influence on Voters. *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 4244–4275. https://doi.org/10.18653/v1/2024.emnlp-main.244

Santurkar, S., Durmus, E., Ladhak, F., Lee, C., Liang, P., & Hashimoto, T. (2023). Whose opinions do language models reflect? In A. Krause, E. Brunskill, K. Cho, B. Engelhardt, S. Sabato, & J. Scarlett (Eds.), *Proceedings of the 40th international conference on machine learning* (pp. 29971–30004, Vol. 202). PMLR.

Tigges, C., Hollinsworth, O. J., Geiger, A., & Nanda, N. (2023). Linear Representations of Sentiment in Large Language Models. https://doi.org/10.48550/ARXIV.2310.15154

Törnberg, P., Valeeva, D., Uitermark, J., & Bail, C. (2023). Simulating Social Media Using Large Language Models to Evaluate Alternative News Feed Algorithms. https://doi.org/10.48550/ARXIV.2310.05984

Vafa, K., Naidu, S., & Blei, D. (2020). Text-Based Ideal Points. *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 5345–5357. https://doi.org/10.18653/v1/2020.acl-main.475

Wu, P. Y., Nagler, J., Tucker, J. A., & Messing, S. (2023). Large Language Models Can Be Used to Estimate the Latent Positions of Politicians. https://doi.org/10.48550/ARXIV.2303.12057